

Supplementary Notes to LVC v12: Form and Lock Degeneracy under DR2

LVC v12 — Supplement

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Note

This is a short supplement to LVC v12 (April 2026) recording four post-publication checks. None of the v12 quantitative conclusions change. The content is provided as reference; the points raised are limitations rather than findings.

1. Functional form

The v12 ansatz $T(z) = 1 + A \exp[-((z - z_c)/w)^2]$ uses a Gaussian profile. Replacing the profile $\exp(-x^2)$ by other localized forms (same parameter count $k = 5$, same DR2 dataset) produces:

Profile $f(x)$	χ^2	$\Delta\chi^2$ vs Gaussian
Gaussian $\exp(-x^2)$	14.48	-
Quartic $\exp(-x^4)$	13.90	-0.58
Lorentzian $1/(1 + x^2)$	14.10	-0.38
$\text{sech}^2 1/\cosh^2(x)$	14.47	-0.01
Exp $\exp(- x)$	14.50	+0.02
Cosine bump	14.30	-0.18
Linear bump	14.48	0.00

All forms cluster in $\chi^2 \in [13.9, 14.5]$. No form is rejected; none is uniquely preferred at $\Delta\chi^2 \geq 1$. The v12 Gaussian is not identified by these data as the unique correct shape.

2. Lock candidates within 1σ

The v12 lock $(z_c, w) = (1/\sqrt{3} + 1/(3\pi), 2/(3\pi)) = (0.6834, 0.2122)$ sits within the DR2 1σ region ($z_c \in [0.636, 0.722]$, $w \in [0.146, 0.270]$). Several other natural-

constant candidates also fall within this region:

For z_c : $\ln 2 = 0.6931$ (0.34σ), $e/4 = 0.6796$ (0.02σ), $1/\sqrt{2} = 0.7071$ (0.66σ), $2/3 = 0.6667$ (0.28σ).

For w : $\pi/15 = 0.2094$ (0.02σ), $e/13 = 0.2091$ (0.01σ), $1/\sqrt{22} = 0.2132$ (0.07σ), $1/5 = 0.2000$ (0.13σ).

The v12 lock is one option among several that fit equally well. With the quartic form (§1), the combination $(\ln 2, 2/(3\pi))$ produces $\chi^2 = 13.90$, marginally lower than v12's 14.50 but well within statistical noise. The data do not pin a unique lock.

3. Continuous family ridge

The v12 family is

$$z_c(n) = \frac{1}{\sqrt{3}} + n \cdot \frac{1}{3\pi}, \quad w(n) = (n+1) \cdot \frac{1}{3\pi}.$$

Treating n as continuous and minimizing on DR2 along this ridge:

n	z_c	w	DR2 χ^2
0.00	0.5774	0.1061	20.62
0.50	0.6304	0.1591	15.54
0.90	0.6728	0.2016	14.51
0.95	0.6782	0.2069	14.49
1.00	0.6834	0.2122	14.50
1.05	0.6888	0.2175	14.55
1.50	0.7365	0.2653	16.40
2.00	0.7896	0.3183	18.71

The continuous minimum is at $n = 0.95$ with $\chi^2 = 14.49$. The integer $n = 1$ value, $\chi^2 = 14.50$, lies 0.013 above the continuous minimum. The v12 family rung is consistent with the continuous best to within negligible tolerance.

4. DR2 leave-one-out

Free 5-parameter refit on DR2 with each individual BAO point removed:

Removed point	free z_c	free w	χ^2
BGS (0.295)	0.670	0.223	13.67
LRG1 (0.510)	0.703	0.186	12.54
LRG2 (0.706)	0.601	2.000 (boundary)	13.42
LRG3+ELG1 (0.934)	0.050 (boundary)	2.000 (boundary)	15.27
ELG2 (1.321)	0.675	0.211	13.19

Removed point	free z_c	free w	χ^2
QSO (1.484)	0.080 (boundary)	2.000 (boundary)	17.14
Ly α (2.330)	0.677	0.213	10.19

Four of seven LOO refits return parameters close to the full-data best-fit ($z_c \approx 0.68$, $w \approx 0.21$). Three refits (LRG2, LRG3+ELG1, QSO removed) have free parameters that escape to boundary values — in those subsets the localized-bump structure is not preferred over a flat or wide modulation. The v12 lock therefore depends on the simultaneous presence of the LRG2, LRG3+ELG1, and QSO measurements at $z \in [0.7, 1.5]$.

5. Residual diagnostics

For the v12 best-fit (DR2 + n=1 lock geometry only), the standardized residuals across all $N = 37$ data points have:

- mean -0.086
- standard deviation 0.78
- $|r| < 1$: 33/37 (89%, ideal $\sim 68\%$)
- $|r| < 2$: 34/37 (92%, ideal $\sim 95\%$)
- $\max |r| = 2.26$

The standard deviation below unity (0.78 vs ideal 1.0) suggests the published BAO covariances may be conservatively inflated, or that the model absorbs more variance than it should. The reduced $\chi^2 = 14.5/(37 - 3) = 0.43$ similarly indicates an over-determined fit by standard convention. Both v12 and Λ CDM share the same data and covariances, so the relative comparison (ΔBIC) is unaffected; the absolute χ^2 values are likely scaled.

Summary of supplement

None of the four checks change v12’s headline numbers: $\Delta\text{BIC} = -11.57$ (geometry only) and -11.25 (joint with $f\sigma_8$). The checks reveal:

- the functional form is not uniquely determined ($\leq 1\sigma$ degeneracy among Gaussian, quartic, Lorentzian, sech^2);
- the lock values are not uniquely determined (multiple natural-constant candidates within 1σ);
- the continuous family minimum sits at $n = 0.95$, marginally below integer $n = 1$ used as the v12 lock;
- the lock structure depends on the joint presence of three intermediate-redshift BAO measurements;
- residuals indicate the absolute χ^2 is plausibly inflated by conservative covariances, but the relative comparison stands.

These are recorded for reference. v12’s quantitative claims are robust; v12’s parameterization is one of several that fit the data equally well.